# AN EXPLICIT FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

Abstract. A countable discrete group is MF here if it embeds into the unitary group of a norm matrix corona. Equivalently, it embeds into the metric quotient of a product of finite-dimensional unitary groups by the sequences converging to the identity in operator norm. We construct an explicit finitely presented group $E$, together with a designated central involution $w \in E$, such that $w \neq 1$ but *every* homomorphism from $E$ to the unitary group of any norm matrix corona sends $w$ to 1. Thus $E$ is not MF, and neither its concrete universe-relative maximal nor its reduced group $C^*$-algebra is MF.

The eight-generator presentation contains a literal six-generator base and a stable letter that doubles its three commuting translation generators. Property (T) of that exact base is certified by a rational degree-one Hodge decomposition for a six-generator, thirteen-relator group, a checked quotient onto the rotation subgroup, and an intrinsic fixed-point argument for the translation subgroup. The obstruction then uses coordinatewise averaging and compression, polar correction, a finite-stage spectral projection, and equality of finite ranks. In general, for the resulting one-sided compression pattern, every finite normal subgroup $F$ inside the compression-defect normal closure lies in the kernel of every corona representation. Two consequences are that $C^*_{\mathrm{red}}(E)$ is stably finite but not MF, and that MF is not preserved by quotients. An explicit cyclic-base comparison shows that exact finite-dimensional invisibility alone does not force the marked word to be trivial. The MF conclusion is a separate uniform statement; here the property-(T) step supplies that additional control.

## 1. Introduction

Let $(d_n)_{n\in\mathbb{N}}$ be a sequence of positive integers and let

$$\mathcal{Q} \;=\; \prod_n M_{d_n}(\mathbb{C}) \Big/ \bigoplus_n M_{d_n}(\mathbb{C}) \tag{1}$$

be the associated *norm matrix corona*: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable $C^*$-algebra is *MF* if it embeds into some algebra of the form (1). We use *MF group* exclusively in the sense introduced by Carrión–Dadarlat–Eckhardt and used by Shulman [CDE, Sh]: a countable discrete group $E$ is MF if, for some strictly increasing sequence of positive integers $(d_n)$, there is an injective homomorphism

$$E \longrightarrow \mathcal{U}(\mathcal{Q}).$$

We shall use the equivalent coordinate model

$$\mathsf{U}_{\mathrm{cor}}((d_n)) := \prod_n \mathcal{U}(M_{d_n}(\mathbb{C}))\Big/\{(u_n) : \|u_n - 1\| \longrightarrow 0\}. \tag{2}$$

The denominator is normal, and polar correction gives a canonical isomorphism

$$\kappa_{(d_n)} \colon \mathsf{U}_{\mathrm{cor}}((d_n)) \xrightarrow{\cong} \mathcal{U}(\mathcal{Q}), \tag{3}$$

proved in Lemma 6.2. Thus one may equivalently embed $E$ into $\mathsf{U}_{\mathrm{cor}}((d_n))$.

**Proposition 1.1** (equivalent MF formulations)**.** *For every countable group, the genuine $C^*$-corona definition above is equivalent to the unitary-sequence definition. The latter is equivalent to operator-norm local models with separation constant $1$, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

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For the final equivalence, place the $n$th model in the last block of dimension $d_0 + \cdots + d_n$ and fill the earlier blocks with identities. This preserves every defect and distance from the identity. Hereafter "MF" means this equivalent actual-corona/local-operator-norm property. It should not be confused with the stronger convention based on reduced-norm convergence of asymptotic representations [Sc, GKEMP], or with strongly PMF, where the approximants are genuine finite-dimensional representations [MdlS, Definition 1.2]. These are genuinely different conventions; every assertion below uses only this MF property. The following theorem gives an explicit finitely presented non-MF group.

**Theorem A** (non-MF group)**.** *Let $E$ be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the finite list of relators in Definition 3.1, and put*

$$w \ := \ \big[\, tct^{-1},\ v_1(tct^{-1})v_1^{-1} \big].$$

*The defining relations make $w$ central and give $w^2 = 1$. Then:*

*(1) $w \neq 1$ in $E$, so $w$ is a central involution;*

*(2) for every sequence $(d_n)$ of positive integers and every group homomorphism $\Theta \colon E \to \mathsf{U}_{\mathrm{cor}}((d_n))$ into the corresponding unitary norm corona (2), one has $\Theta(w) = 1$.*

*In particular $E$ is not MF, and neither the universe-relative algebra $C^*_{\max}(E)$ fixed below nor $C^*_{\mathrm{red}}(E)$ is an MF $C^*$-algebra.*

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The inline label links the theorem directly to its declaration. Similar labels below give the declaration name for each numbered claim.

A completely explicit presentation of $E$ — eight generators, with every relation displayed — appears as (7) in Definition 3.1.

The proof is coordinatewise and has five steps. Let $F \trianglelefteq H$ be the finite normal subgroup in the general criterion; for Theorem A, $F = \{1, w\}$.

(1) *Finite-group averaging.* From a coordinate model nontrivial on a finite normal subgroup $F$, form the coordinate averages $A_n = |F|^{-1} \sum_{f \in F} U_{f,n}$. Hermitianize and spectrally round them, then retain the complementary nonzero spectral blocks.

(2) *Compression and polar repair.* Normality of $F$ makes the rounded projections asymptotically central. Principal compression therefore preserves multiplication up to vanishing operator-norm error; its Gram defects vanish, and polar correction gives honest unitary matrices without changing the asymptotics (Lemma 7.3).

(3) *Vanishing finite sum.* On the complementary blocks the Reynolds averages vanish asymptotically. Consequently the repaired matrices satisfy

$$\left\| \sum_{f \in F} V_{f,n} \right\| \longrightarrow 0.$$

(4) *Property-*(T) *spectral projection.* A finite-stage spectral projection for the adjoint microstates of $\Gamma$ is almost contained in its $t$-translate. The two projections have equal rank, so the complementary block estimate reverses the containment. This forces every generator $[tct^{-1}, \iota(\lambda)]$ to be Hilbert–Schmidt trivial (Lemma 7.4).

(5) *Contradiction.* In a tracial matrix ultraproduct the whole normal closure, hence $F$, is trivial. A cofinal diagonal extraction makes every $V_{f,n}$ Hilbert–Schmidt close to $V_{1,n}$, contradicting the vanishing finite sum from step (3) (Section 8).

Figure 1 shows where the two halves of the proof meet. The upper lane is purely group-theoretic; the lower lane is the finite-matrix rigidity input. Their common output is incompatible with the complementary finite-group corner.

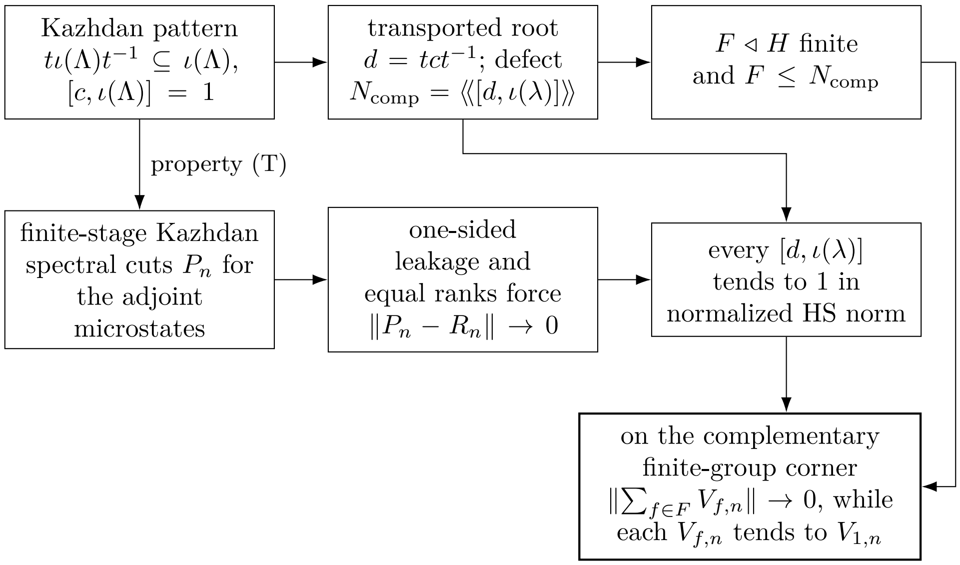


FIGURE 1. The finite-normal obstruction. The upper lane is group-theoretic and the lower lane is the finite-matrix rigidity input. Property (T) supplies a uniform spectral cut, and equality of finite ranks turns its one-sided leakage estimate into two-sided control.

Nontriviality of $w$ is witnessed by a completely explicit representation of $E$ into a semidirect product of a Clifford group by a linear group over $\mathbb{Z}[1/2]$ (Section 4); no combinatorial theory of HNN extensions is needed.

Two companion results clarify the mechanism. First, in finite dimensions the obstruction is elementary and needs neither property (T), nor unitarity, nor centrality of $w$:

For a subgroup $L \leq H$, put

$$\mathcal{C}(L) = \{s \in H : sLs^{-1} \subseteq L\}, \qquad G_{\mathrm{comp}}(L) = \langle \mathcal{C}(L) \rangle,$$

and define the intrinsic *compression–centralizer defect*

$$\mathfrak{D}(H, L) = \left\langle\!\!\left\langle [gzg^{-1}, \ell] : g \in G_{\mathrm{comp}}(L),\ z \in C_H(L),\ \ell \in L \right\rangle\!\!\right\rangle_H . \tag{4}$$

**Theorem B** (finite-dimensional obstruction)**.** *Let $k$ be a field, $V$ a finite-dimensional $k$-vector space, $H$ a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi \colon H \to \mathrm{GL}(V)$ satisfies*

$$\pi\big([\, tct^{-1},\ a(tct^{-1})a^{-1}]\big) \ = \ 1.$$

*More generally, every such $\pi$ sends the entire subgroup $\mathfrak{D}(H, \Gamma)$ to the identity.*

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Applied to $H = E$, Theorem B shows that *every* finite-dimensional representation of $E$ over *every* field sends $w$ to the identity: finite-dimensional linear representations do not separate the points of $E$, no finite quotient can witness $w \neq 1$, and the infinite Clifford witness of Section 4 cannot be replaced by a finite-dimensional one. Property (T) enters the corona argument for one reason: it makes "the projection onto the $\Gamma$-fixed space" an element of the corona, uniformly across all coordinates at once.

Second, the elementary finite-dimensional calculation does not determine whether its marked word is trivial:

**Theorem C** (cyclic-base comparison)**.** *Define the literal marked presentation*

$$E_{\mathrm{BS}} = \left\langle \gamma_0, t, c \;\middle|\; \begin{matrix} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\mathrm{BS}}^2 = 1, \quad [w_{\mathrm{BS}}, \gamma_0] = [w_{\mathrm{BS}}, t] = [w_{\mathrm{BS}}, c] = 1 \end{matrix} \right\rangle,$$

*where* $w_{\mathrm{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$*. Then* $E_{\mathrm{BS}}$ *is finitely presented and* $w_{\mathrm{BS}} \neq 1$*. There is an explicit surjective homomorphism from* $E_{\mathrm{BS}}$ *onto its realized Clifford quotient that maps* $w_{\mathrm{BS}}$ *to the nontrivial central sign. Nevertheless every finite-dimensional representation of* $E_{\mathrm{BS}}$ *over every field maps* $w_{\mathrm{BS}}$ *to the identity.*

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The quotient statement makes the comparison at the level actually used by the proof: exact finite-dimensional invisibility does not by itself force triviality of the marked word. The MF conclusion in Theorem A is the separate uniform assertion supplied by the fixed-space argument.

**Generalizations and consequences.** Theorem A is an instance of a general finite-normal obstruction (Definition 7.1). A marked Kazhdan pattern consists of a property (T) group $\Lambda$, a homomorphism $\iota\colon \Lambda \to H$, and elements $t, c \in H$ such that $t\iota(\Lambda)t^{-1} \subseteq \iota(\Lambda)$ and $[c, \iota(\Lambda)] = 1$. Theorem 7.2 sends every finite normal subgroup contained in the resulting compression-defect subgroup to the identity under every corona representation. For $E$, the relevant subgroup is $\{1, w\}$. Corollary 9.5 records the resulting obstruction for $C^*$-subalgebras of norm matrix coronas, while Theorem C shows that finite-dimensional collapse by itself does not decide whether the marked word is trivial.

Section 9 develops structural and $C^*$-algebraic consequences. The principal $C^*$-algebraic application is:

**Theorem D** (a stably finite non-MF reduced group algebra)**.** $C^*_{\mathrm{red}}(E)$ *is a unital separable* $C^*$*-algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

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**What this paper does not prove.** The proof of Theorem A does *not* show that $E$ is nonhyperlinear. The finite-rank step in the proof is specific to operator norm: nested unitarily equivalent projections need not coincide in the properly infinite operator algebra arising from a tracial ultraproduct. Section 11 explains this failure. The argument also leaves $E/\langle w\rangle$ unresolved.

**External inputs.** The load-bearing property-(T) statement is proved for the literal twenty-relator base in Section 3; it is not imported from an identification of that presentation with a semantic matrix group. The proof uses a finite rational Hodge certificate, checked relator replays, and the Hilbert-space fixed-point theorem. The Clifford witness and the finite-dimensional obstruction are proved directly.

**Relation to prior work.** The closest conceptual analogue is the recent work on nonsofic groups. One-sided compression of a property (T) subgroup and propagation from its centralizer appear in [OAI, KT], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; related centralizer rigidity for sofic embeddings is proved in [AT]. Those results concern Hamming or tracial approximations. Here coordinatewise averaging, polar correction, and a finite-stage spectral projection produce a specific element in the kernel of every operator-norm corona representation. The Clifford central extension supplies the element whose survival is proved independently.

The operator-norm argument replaces finite cardinality by the implication

$$\text{one-sided leakage of } P_n \text{ into } R_n, \qquad \operatorname{rank} P_n = \operatorname{rank} R_n \\ \Longrightarrow \qquad \|P_n - R_n\| \longrightarrow 0.$$

The complementary finite-group corner then converts this estimate into a fixed nontrivial element sent to the identity by every corona representation.

Slofstra's constructions supply close precedents for the Clifford group, distinguished central involution, shift, and HNN doubling map [Slo18, Section 2 and Remark 3.3]. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedure in detail [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]. The present proof combines that local corner procedure with the one-sided Kazhdan compressor; equality of the two finite ranks supplies the reverse projection estimate and hence the universal kernel conclusion.

The analytic passage from operator norm to the Hilbert–Schmidt conjugation representation belongs to the $\pi \otimes \bar{\pi}$ framework of Bekka and Bekka–Valette [Be, BV]. Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. These are the direct analytic precedents for the conjugation step used here.

Slofstra's quotients by elements invisible to exact or approximate finite-dimensional representations precede the MF-radical language used below

[Slo17, Definitions 2.5–2.7]. Fritz's marked BS(2, 3) example provides additional context for the cyclic comparison [Fri, SV].

Central extensions and rigidity of approximate representations also drive the Frobenius obstruction of De Chiffre–Glebsky–Lubotzky–Thom [DGLT]. In the proof of Proposition 1.6, Bachner–Dogon–Lubotzky treat the finite-central case by a character-isotypic corner, and reduce the general finite-normal case to it by passing to the finite-index centralizer [BDL]. The finite-normal criterion here combines the corresponding coordinate-corner construction with the one-sided Kazhdan compressor and does not assume that the quotient is hyperlinear. Related finite-extension formulations of soficity appear in [GR, Gl].

Carrión–Dadarlat–Eckhardt provide the group-MF framework used here and, through Abels' group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. The cyclic comparison addresses the narrower question of what happens to the marked compression pattern when property (T) is removed.

Finally, failure of the MF property for a group $C^*$-algebra does not by itself prevent the group from embedding into a corona: the group property requires only a faithful canonical image in some MF quotient of $C^*_{\max}(E)$. The result here produces a specific nontrivial element in the kernel of every such representation. For broader context see [DGLT, BDL, OAI, KT, Sh, Dad, Slo17, Slo18, SV].

**Organization.** Section 2 fixes notation and states the external inputs. Section 3 constructs $E$ and certifies property (T) for its literal base. Section 4 proves $w \neq 1$. Section 5 proves Theorem B. Section 6 develops the corona lemmas. Section 7 defines marked Kazhdan patterns and their compression-defect subgroups, states the finite-normal obstruction criterion (Theorem 7.2), and develops the coordinate Reynolds and Kazhdan-compression estimates. Section 8 proves the criterion and deduces Theorem A. Section 9 derives Theorem D and further consequences. Section 10 proves Theorem C. Section 11 discusses limitations.

## 2. Notation and external inputs

All groups are discrete and countable, and every group and auxiliary index set is represented on a fixed small carrier. This is no restriction for the objects considered here: a countable group is isomorphic to one whose underlying set is a subset of $\mathbb{N}$, and finite coordinate systems can be reindexed by natural numbers. Property (T) below is the textbook complex-unitary property, independent of the universe containing the Hilbert space. A $C^*$-homomorphism is unital only when this is stated. In particular, an MF embedding may be nonunital, as is customary for embedding a unital algebra as a $C^*$-subalgebra. If $e\colon A \to B$ is an injective, possibly nonunital $*$-homomorphism between unital $C^*$-algebras, then $u \mapsto e(u) + (1 - e(1))$ is an injective homomorphism $\mathcal{U}(A) \to \mathcal{U}(B)$; this complement correction is used below. For a unital $C^*$-algebra $A$ we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace $\mathrm{tr}_r$, the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \mathrm{tr}_r(x^*x)^{1/2}$. We will use three elementary facts without further comment: $\|x\|_2 \leq \|x\|$; $|\,\mathrm{tr}_r(x)| \leq \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries $u, v$.

Given a sequence $(d_n)_{n\in\mathbb{N}}$ of positive integers, we write

$$\mathcal{Q}\big((d_n)_{n\in\mathbb{N}}\big) \;=\; \prod_{n\in\mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) \Big/ \bigoplus_{n\in\mathbb{N}}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded $C^*$-product by the ideal of sequences with $\|x_n\| \to 0$. We call any such algebra a *norm matrix corona*. We also write

$$\mathsf{U}_{\mathrm{cor}}((d_n)_{n\in\mathbb{N}}) = \prod_{n\in\mathbb{N}} \mathcal{U}(M_{d_n}(\mathbb{C})) \Big/ \{(u_n) : \|u_n - 1\| \longrightarrow 0\}.$$

A *corona representation* of a group $E$ is a homomorphism $E \to \mathcal{U}(\mathcal{Q})$. Lemma 6.2 canonically identifies it with a homomorphism into $\mathsf{U}_{\mathrm{cor}}((d_n))$, which we call its coordinate model. The group $E$ is *MF* if it admits an injective corona representation for a strictly increasing dimension sequence. By the dimension-amplification equivalence discussed in the introduction, allowing an arbitrary positive dimension sequence gives the same class of groups.

Fix a universe containing every group and $C^*$-algebra used below. For a group $H$ in it, $C^*_{\max}(H)$ denotes the following concrete universe-relative maximal model. Take the bounded product of every same-universe nontrivial unital $C^*$-algebra equipped with a homomorphism $H \to \mathcal{U}(B)$, map $H$ diagonally into this product, and take the closed $*$-subalgebra generated by that image. We write $u_h$ for its canonical unitaries.

**Proposition 2.1** (universe-relative maximal model)**.** *The canonical map $h \mapsto u_h$ is injective. For every same-universe unital $C^*$-algebra $B$ and every homomorphism $\rho\colon H \to \mathcal{U}(B)$, there is a unique unital $*$-homomorphism $C^*_{\max}(H) \to B$ that maps $u_h$ to $\rho(h)$ for every $h \in H$.*

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`GroupApproximation.manuscriptUniverseRelativeMaximalGroupCStar`

Coordinate evaluation gives the extension, while the left-regular coordinate proves injectivity. These are precisely the two properties of the maximal construction used here.

We use the commutator convention $[x, y] = xyx^{-1}y^{-1}$.

## 3. The literal group and its Kazhdan certificate

We work throughout with the groups defined by the relators below. No identification of these presented groups with a matrix group is used. First define the *rotation group*

$$\mathcal{R} = \left\langle x, y, z \;\middle|\; \begin{matrix} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{matrix} \right\rangle, \tag{5}$$

and the *literal base*

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \;\middle|\; \begin{matrix} \text{the eight relations in (5)}, \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3,\ xv_2x^{-1} = v_1,\ xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1,\ yv_2y^{-1} = v_2^{-1}v_3,\ yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1},\ zv_2z^{-1} = v_1v_3^{-1},\ zv_3z^{-1} = v_3^{-1} \end{matrix} \right\rangle. \tag{6}$$

Write $\iota\colon \mathcal{B} \to E$ for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

**Definition 3.1.** Use the six literal base generators and the relations displayed in (6). Define

$$E = \left\langle v_1, v_2, v_3, x, y, z, t, c \;\middle|\; \begin{matrix} \text{all relations in (6)}, \\ tv_it^{-1} = v_i^2\ (1 \le i \le 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \qquad w^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{matrix} \right\rangle, \tag{7}$$

where $w$ abbreviates the word $\left[\, tct^{-1},\, v_1(tct^{-1})v_1^{-1}\right]$. This is an explicit finite presentation with eight generators. Its defining relations make $w$ central and impose $w^2 = 1$; the relations on the six base generators also give

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \qquad [c, \iota(\mathcal{B})] = 1.$$

Proposition 4.3 below proves that $w \neq 1$.

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Figure 2 isolates the geometry used later in the analytic argument: $d = tct^{-1}$ centralizes the compressed copy, not necessarily all of $\iota(\mathcal{B})$, and the marked commutator measures precisely that failure after conjugating $d$ by $\iota(v_1)$.

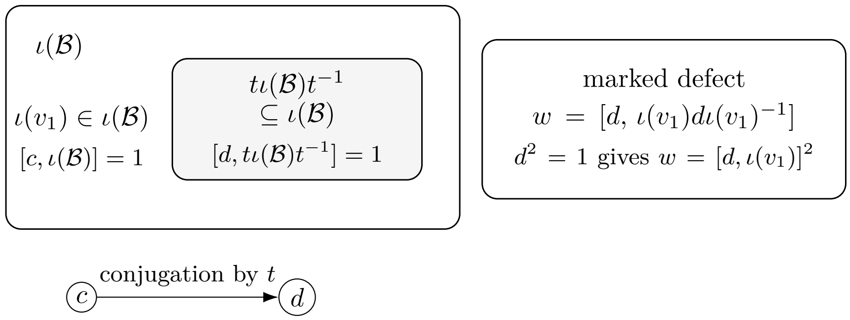


FIGURE 2. The marked-compression pattern. Conjugation by $t$ carries the image of the literal base into itself. It sends $c$ to $d = tct^{-1}$, which centralizes that compressed copy. Only the one-sided inclusion is used; no injectivity or strict containment is asserted.

The presentation itself, rather than an asserted identification with a semantic matrix group, is the source of every relation used below. In particular the argument needs only the one-sided inclusion and centralizer relations shown in Figure 2; it never uses injectivity or properness of the base map.

### 3.1. **An exact property-(T) certificate.**

We next prove property (T) for the exact group $\mathcal{B}$.

**Proposition 3.2** (exact rational Hodge certificate)**.** *Let $P_{13}$ be the group on the six ordered generators*

$$e_{12}, e_{13}, e_{21}, e_{23}, e_{31}, e_{32}.$$

*Its first six relators are*

$$[e_{12}, e_{23}]e_{13}^{-1}, \quad [e_{13}, e_{32}]e_{12}^{-1}, \quad [e_{21}, e_{13}]e_{23}^{-1},$$
$$[e_{23}, e_{31}]e_{21}^{-1}, \quad [e_{31}, e_{12}]e_{32}^{-1}, \quad [e_{32}, e_{21}]e_{31}^{-1};$$

*the next six are the commutators of the displayed commuting-root pairs*

$$(12, 13),\ (12, 32),\ (13, 23),\ (21, 23),\ (21, 31),\ (31, 32),$$

*and the last is $(e_{12}e_{21}^{-1}e_{12})^4$.*

*Put $D_i = e_i - 1 \in \mathbb{Q}[P_{13}]$, and let $B_1$ be the $13 \times 6$ Fox-boundary matrix of the displayed relators. Fox telescoping gives $B_1 D = 0$. A finite certificate, whose entries are exact rational data, gives a matrix $Q \in M_{102\times 6}(\mathbb{Q}[P_{13}])$ supported on 22 explicit words in the certificate, with denominator $10^8$, and $R \in M_6(\mathbb{Q}[P_{13}])$ such that*

$$B_1^* B_1 + DD^* = \frac{1}{250} I_6 + Q^* Q + R. \tag{8}$$

*The 22-by-22 support products are accompanied by explicit relator replays; they are collected into 293 finite coordinates. Both orientations of the residual estimate are checked:*

$$\max_i \sum_k \|R_{ik}\|_1, \quad \max_k \sum_i \|R_{ik}\|_1 \;\le\; \rho := \frac{11670886519714}{10^{16}} < \frac{1}{250}. \tag{9}$$

*Thus the certified gap is*

$$\gamma = \frac{1}{250} - \rho = \frac{28329113480286}{10^{16}} > 0. \tag{10}$$

*Let $\varepsilon_{13} > 0$ be the explicit certified tolerance obtained from these rational data by the formula in the exact declaration. Then the six displayed generators, with tolerance $\varepsilon_{13}$, form a Kazhdan pair for $P_{13}$. Consequently $P_{13}$ has property* (T) *in both the real-orthogonal and textbook complex-unitary formulations, in every representation-space universe.*

$\boxed{\textsf{EXACT}}$

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*Proof.* The relator replay, Fox identity, coefficient decomposition, and rational row and column bounds are the finite checks recorded in the proposition. The 293 coordinates need not be distinct in the abstract group: collecting additional collisions can only decrease the true group-ring coefficient $\ell^1$-norm, so the checked coordinate sums give valid upper bounds. For the analytic passage, let $\pi$ be an orthogonal representation on a real Hilbert space and put

$$A = \sum_i (\pi(e_i) - 1)^*(\pi(e_i) - 1), \qquad z_i = (\pi(e_i) - 1)\xi.$$

The identity $B_1 D = 0$ and (8) give

$$\gamma\langle A\xi, \xi\rangle \le \|A\xi\|^2.$$

With $M = \max\{\gamma, 24\}$ and

$$\sigma = \gamma(1 - \sqrt{1 - \gamma/M}),$$

the usual self-adjoint gap estimate gives $\sigma\|\xi\| \le \|A\xi\|$ on the subspace without invariant vectors. Since $\|A\xi\| \le 2\sum_i \|\pi(e_i)\xi - \xi\|$, the six generators form a Kazhdan pair with tolerance $\sigma/12$, which is the certified value $\varepsilon_{13}$. Realification and complexification identify the real and textbook complex formulations. □

Figure 3 records the scale of the exact certificate. Each heat-map cell represents an upper bound for the coefficient $\ell^1$-mass of one block of $R$; both its row and column sums are checked.

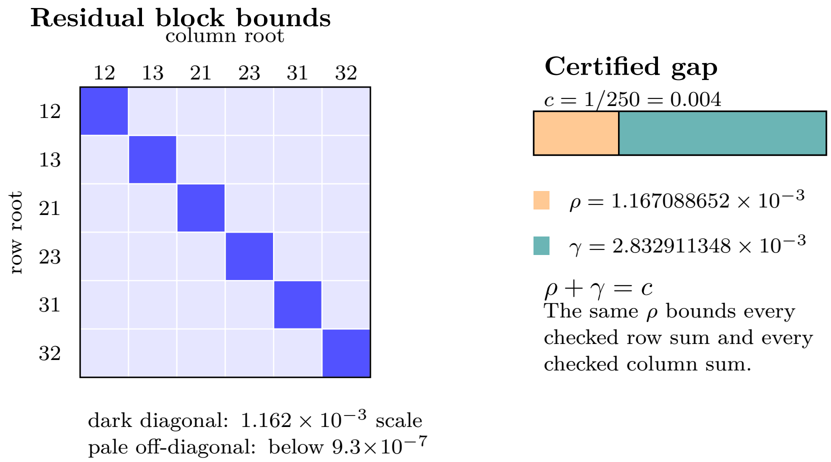


FIGURE 3. Exact Hodge-certificate scale. The residual grid distinguishes the diagonal and off-diagonal block bounds; the separate bar partitions $c = 1/250$ into the certified residual radius $\rho$ and positive gap $\gamma = c - \rho$.

To pass from $P_{13}$ to the literal base, substitute

$$e_{12} = y^{-1}xzy^{-1}x^{-1}, \quad e_{13} = xzy^{-1}x^{-1}y^{-1}, \quad e_{21} = y^{-1}xyxz,$$
$$e_{23} = x^{-1}y^{-1}xzy^{-1}, \quad e_{31} = xy^{-1}xzy^{-1}x, \quad e_{32} = x^{-1}zy^{-1}x^{-1}y^{-1}x^{-1}.$$

Every substituted $P_{13}$ relator is a checked product of conjugates of the eight relators of $\mathcal{R}$. The words

$$e_{12}e_{21}^{-1}e_{31}^{-1}e_{12}e_{13}e_{31}^{-2}e_{13}, \qquad e_{12}^{-1}e_{13}^{-1}e_{31}$$

map respectively to $z$ and $xy$, so the resulting homomorphism $P_{13} \to \mathcal{R}$ contains $z$ and $xy$ in its image. A separate 31-step relator replay in the literal eight-relator presentation proves $\mathcal{R} = \langle z, xy \rangle$: it writes $y$ as a word in $z$ and $xy$, after which $x = (xy)y^{-1}$. Hence the homomorphism is surjective, and property (T) descends to $\mathcal{R}$.

Finally let $(S, \kappa)$ be a Kazhdan pair for $\mathcal{R}$, with $\kappa \leq 1$. For the image of $S \cup \{v_2, v_3\}$ in $\mathcal{B}$, use tolerance $\kappa/64$. Project an almost invariant unit vector to an $\mathcal{R}$-fixed vector $p$ within $1/64$. The literal conjugation relations express every translation as a product of two $\mathcal{R}$-conjugates; hence every translation moves $p$ by at most $1/8$. The bounded-orbit fixed-point theorem bounds the distance from $p$ to the closed translation-fixed subspace by $1/8$. Let $q$ be the orthogonal projection of $p$ onto that subspace. Rotations normalize the translation subgroup and fix $p$; uniqueness and equivariance of orthogonal projection therefore make $q$ rotation-fixed as well. Hence $q$ is fixed by all of $\mathcal{B}$. Moreover $\|q - p\| \leq 1/8$ and $\|p - x\| < 1/64$, so $\|q - x\| < 1$; since $x$ is a unit vector, $q \neq 0$.

**Proposition 3.3.** *The literal twenty-relator group $\mathcal{B}$ has property* (T).

EXACT

GroupApproximation.LiteralBaseP13PropertyTBridge.manuscriptBaseHasKazhdanPropertyT

## 4. The witness: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp only as an explicit algebraic witness; it makes no identification of the literal base with the matrix realization used here.

**Construction 4.1** (Clifford lamp groups)**.** For a set $X$, form the real Clifford algebra on generators $(c_x)_{x\in X}$ and let $\mathsf{ClLamp}(X)$ be its subgroup of units generated by $-1$ and the $c_x$. Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \qquad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign $\zeta$ is nontrivial. Every permutation of $X$ acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes $\zeta$.

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptCliffordConstruction

**Lemma 4.2** (linear model)**.** *For $M \in \mathrm{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M,u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

*Assign $v_i \mapsto A(I_3, e_i)$ and*

$$x \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), \qquad y \mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right),$$

$$z \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right).$$

*Let $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \mathrm{diag}(2,2,2,1)$. Direct matrix computation proves that the six matrices satisfy all twenty relators of $\mathcal{B}$ and that*

$$\bar{\alpha}(u) = DuD^{-1}$$

*is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes $x, y, z$. Moreover the matrix assigned to $v_1$ does not lie in $\mathrm{range}(\bar{\alpha})$.*

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptLinearModel

*Proof.* The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by $D$ gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every generator of $\bar{\Gamma}$ has integral entries, and integrality is preserved under products and inverses; however $D^{-1}\bar{v}_1 D$ has entry $1/2$ in position $(1,4)$. Thus it is not in $\bar{\Gamma}$, which proves the last assertion. □

Form the mapping telescope of $\bar{\alpha}$, extend it by its shift $\tau$, and call the resulting group $V$. Let $j\colon \bar{\Gamma} \hookrightarrow V$ be its level-zero map and put $X = V/j(\bar{\Gamma})$, with root $o = j(\bar{\Gamma})$. The final assertion of Lemma 4.2 is exactly the statement that

$$\tau o \neq j(\bar{v}_1)\tau o.$$

Thus the Clifford generators at these two sites anticommute.

**Proposition 4.3.** *$w \neq 1$ in $E$.*

EXACT

`GroupApproximation.LiteralNonMFLinearWitness.literal_mark_ne_one`

*Proof.* Form $W := \mathsf{ClLamp}(X) \rtimes V$ as in Construction 4.1. Define a map on the generators of (7) by

$$g \mapsto j(\bar{g}) \in V \quad (g \in \{v_1, v_2, v_3, x, y, z\}),$$
$$t \mapsto \tau, \qquad c \mapsto c_o.$$

The exact checks in Lemma 4.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes $o$, so $c_o$ commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \qquad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two sites are distinct by Lemma 4.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since $\zeta$ is a central involution of $W$, the remaining relations $w^2 = 1$ and $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \to W$ sending $w$ to $\zeta \neq 1$, and therefore $w \neq 1$ in $E$. □

Scope of the witness. The target uses the infinite group $\mathsf{ClLamp}(X)$ when $X$ is infinite, but each group element involves only finitely many Clifford sites; the remaining data are rational matrices. No normal-form theorem for an HNN extension enters: the mapping telescope supplies the required group and the range characterization separates the two sites.

## 5. The finite-dimensional obstruction

We prove Theorem B. Everything happens inside the finite-dimensional algebra $\mathrm{End}_k(V)$.

*Proof of Theorem B.* Let $\pi\colon H \to \mathrm{GL}(V)$ be a homomorphism and let

$$C \;=\; \{\, x \in \mathrm{End}_k(V) \;:\; \pi(\gamma)\, x = x\, \pi(\gamma) \text{ for all } \gamma \in \Gamma \,\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\mathrm{End}_k(V)$. Let $u = \pi(t)$ and consider the linear automorphism $\Phi(x) = uxu^{-1}$ of $\mathrm{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and set $y = u^{-1}xu$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with $x$, and therefore

$$\pi(\gamma)\, y = u^{-1}\, \pi(t\gamma t^{-1})\, x\, u = u^{-1}\, x\, \pi(t\gamma t^{-1})\, u = y\, \pi(\gamma).$$

Thus $y \in C$, proving the claim. Since $\Phi$ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$C \;=\; \Phi(C) \;=\; uCu^{-1}. \tag{11}$$

Now $\pi(c) \in C$ because $c$ centralizes $\Gamma$. By (11), $\pi(tct^{-1}) = u\, \pi(c)\, u^{-1} \in C$. Since $a \in \Gamma$, the element $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi\big(a(tct^{-1})a^{-1}\big) = \pi(tct^{-1})$, and therefore

$$\pi\big([\, tct^{-1},\ a(tct^{-1})a^{-1}]\big) = [\, \pi(tct^{-1}),\ \pi(tct^{-1})] = 1. \qquad \square$$

The same calculation proves the final, subgroup-valued assertion of Theorem B. The elements of $H$ whose conjugation action preserves $C$ in both directions form a subgroup. The argument above puts every one-sided compressor in that subgroup, and therefore puts all of $G_{\mathrm{comp}}(\Gamma)$ in it. Thus, for $g \in G_{\mathrm{comp}}(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to $C$ and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence $\pi$ kills all the generators in (4), and normality of its kernel kills their normal closure.

**Corollary 5.1.** *The group $E$ of Definition 3.1 admits no injective homomorphism into $\mathrm{GL}(V)$ for any finite-dimensional vector space $V$ over any field. More strongly, every finite-dimensional linear representation maps $w$ to the identity, and every homomorphism to a finite group maps $w$ to the identity.*

**EXACT**

`GroupApproximation.LiteralFiniteDimensionalObstruction.manuscriptFiniteDimensionalConsequences`

*Proof.* Apply Theorem B with $H = E$, the image of $\mathcal{B}$, and the elements $t, c, v_1$; the hypotheses hold by the defining relations (7). Every finite-dimensional representation kills $w$, and $w \neq 1$ by Proposition 4.3. A finite target embeds into a finite symmetric group and hence into a finite-dimensional general linear group via permutation matrices, so it also maps $w$ to the identity. $\square$

Why the corona argument is different. The proof of Theorem B used only that a finite-dimensional subspace cannot be properly contained in an isomorphic image of itself. An MF model is only asymptotically multiplicative, so its fixed spaces are not literal invariant subspaces. Property (T) repairs this at finite stages: a uniform spectral cut of the adjoint averages captures the almost-fixed vectors. One-sided compression gives a leakage estimate between two equal-rank coordinate projections, and the equal-rank flip supplies the reverse estimate. This is the route used below.

## 6. Norm matrix coronas

Throughout this section $\mathcal{Q} = \mathcal{Q}((d_n)_{n\in\mathbb{N}})$ is a norm matrix corona, $q\colon \prod_{n\in\mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) \to \mathcal{Q}$ the quotient map.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

**Lemma 6.1** (lifting)**.** *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.*

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptUnitaryLifting`

*Proof.* Lift $u$ to an arbitrary bounded sequence $(x_n)$. Since $q((x_n))$ is unitary, both $\|x_n^*x_n - 1\| \to 0$ and $\|x_nx_n^* - 1\| \to 0$. In particular, $x_n^*x_n$ is invertible, and hence $x_n$ is invertible, for all sufficiently large $n$. At those indices put $u_n = x_n(x_n^*x_n)^{-1/2}$. Then $u_n$ is unitary, and, once $\|x_n^*x_n - 1\| \leq \frac{1}{2}$, continuous functional calculus gives the quantitative estimate

$$\|u_n - x_n\| \leq 2\,\|x_n\|\,\|x_n^*x_n - 1\|\,.$$

The sequence $(x_n)$ is bounded, so the right-hand side tends to 0. Setting $u_n = 1$ at the finitely many remaining indices therefore gives a unitary sequence with $\|u_n - x_n\| \to 0$. ☐

**Lemma 6.2** (unitaries in the quotient)**.** *For every sequence $(d_n)$, the map*

$$\kappa_{(d_n)}\colon \mathsf{U}_{\mathrm{cor}}((d_n)) \longrightarrow \mathcal{U}(\mathcal{Q}), \qquad [(u_n)] \longmapsto q((u_n)),$$

*is a canonical group isomorphism.*

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptUnitaryCoronaEquivalence`

*Proof.* The map is well defined, and its kernel consists exactly of the unitary sequences satisfying $\|u_n - 1\| \to 0$, so it is injective. It is surjective by Lemma 6.1: every unitary in $\mathcal{Q}$ has a unitary lift $(u_n)$, whose class in (2) maps to it. ☐

## 7. Kazhdan compression and the finite-normal corner

The corona half of Theorem A will be deduced from a general criterion, whose data we now fix.

**Definition 7.1** (marked Kazhdan pattern)**.** Let $H$ be a countable group. A *marked Kazhdan pattern* in $H$ is a tuple $(\Lambda, \iota, t, c)$, where $\Lambda$ is a countable group with property (T), $\iota\colon \Lambda \to H$ is a group homomorphism — *not* assumed injective — and $t, c \in H$, subject to

(M1) $t\,\iota(\lambda)\,t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;

(M2) $c\,\iota(\lambda) = \iota(\lambda)\,c$ for every $\lambda \in \Lambda$.

Write $d := tct^{-1}$. The *compression-defect subgroup* of the pattern is the normal closure

$$N_{\mathrm{comp}} \;=\; \langle\!\langle\, [\,d,\iota(\lambda)\,] \;:\; \lambda\in\Lambda \,\rangle\!\rangle_H \;\trianglelefteq\; H.$$

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptMarkedKazhdanPattern`

**Theorem 7.2** (finite-normal obstruction criterion)**.** *Let $H$ be a countable group with a marked Kazhdan pattern $(\Lambda,\iota,t,c)$, and let $F\trianglelefteq H$ be a finite normal subgroup with $F\subseteq N_{\mathrm{comp}}$. Then every corona representation $\Theta\colon H\to\mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f)=1$ for all $f\in F$.*

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptFiniteNormalObstructionCriter ion`

Theorem A(2) is the instance $H=E$, $\Lambda=\mathcal{B}$, $\iota$ the canonical homomorphism, and $F=\{1,w\}$. The membership $w\in N_{\mathrm{comp}}$ is the identity $w=[d,\iota(v_1)]^2$ of Lemma 8.1. The proof uses the following coordinate construction. This formulation is important: the Reynolds average is formed before passing to any quotient, and its complementary spectral corner is repaired coordinate by coordinate.

**Lemma 7.3** (coordinate finite-normal corner)**.** *Let $F\trianglelefteq H$ be finite, let $\omega$ be an ultrafilter on an index set $I$, and let $\rho$ be a homomorphism from $H$ into the operator-norm metric ultraproduct of the groups $\mathcal{U}(M_{d_i}(\mathbb{C}))$, $i\in I$. If $\rho|_F$ is nontrivial, then there are nonempty finite coordinate sets $Y_n$ and unitaries $V_{g,n}\in\mathcal{U}(M_{Y_n}(\mathbb{C}))$ such that*

$$\|V_{g,n}V_{h,n}-V_{gh,n}\|\longrightarrow 0\quad (g,h\in H),\qquad \left\|\sum_{f\in F}V_{f,n}\right\|\longrightarrow 0.$$

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptCoordinateFiniteNormalCorner`

*Proof.* Choose $f_0\in F$ with $\rho(f_0)\neq 1$. A standard sequential extraction from the ultraproduct gives representatives $U_{g,n}$ which are asymptotically multiplicative in operator norm and a constant $\delta>0$ such that

$$\|U_{f_0,n}-U_{1,n}\|\geq\delta$$

for every $n$. We now construct the corner model from these coordinates. Put

$$A_n=\frac{1}{|F|}\sum_{f\in F}U_{f,n},\qquad H_n=\frac{1}{2}(A_n+A_n^*).$$

Left and right translation permute $F$, while conjugation by any $g\in H$ permutes $F$ by normality. Expanding the finitely many multiplicative defects therefore shows, in operator norm, that

$$A_n^2-A_n\longrightarrow 0,\qquad A_n-H_n\longrightarrow 0,\qquad U_{g,n}A_nU_{g,n}^*-A_n\longrightarrow 0.$$

Let $P_n = \chi_{(1/2,\infty)}(H_n)$ and $Q_n = 1 - P_n$. Spectral rounding gives

$$\|P_n - H_n\| \le 2 \left\| H_n^2 - H_n \right\| \longrightarrow 0,$$

so $P_n - A_n \to 0$ and $U_{g,n} P_n U_{g,n}^* - P_n \to 0$ for every fixed $g$. The separation of $f_0$ forces $Q_n \neq 0$ eventually: if $Q_n = 0$, then $P_n = 1$, and the relations $U_{f_0,n} A_n - A_n \to 0$, $P_n - A_n \to 0$, and $U_{1,n} - 1 \to 0$ would imply $\|U_{f_0,n} - U_{1,n}\| < \delta$.

Conjugate into an eigenbasis of $H_n$ and let $C_{g,n}$ be the principal block of $U_{g,n}$ on $Q_n \mathbb{C}^{d_n}$; put $r_n = \operatorname{rank} Q_n > 0$. The preceding commutator estimate makes both off-diagonal blocks tend to zero. Hence

$$\|C_{g,n} C_{h,n} - C_{gh,n}\| \longrightarrow 0, \qquad \left\| C_{g,n}^* C_{g,n} - 1 \right\| \longrightarrow 0.$$

For each fixed $g$ and all sufficiently large $n$ the compressed Gram matrix is invertible; take the polar correction

$$V_{g,n} = C_{g,n} (C_{g,n}^* C_{g,n})^{-1/2}.$$

When the Gram defect is at most $1/2$, functional calculus gives the dimension-free estimate

$$\|V_{g,n} - C_{g,n}\| \le 2 \left\| C_{g,n}^* C_{g,n} - 1 \right\|,$$

For each fixed $g$, use this polar unitary on the cofinite tail and set $V_{g,n} = 1$ at its finitely many exceptional indices. Thus polar correction is asymptotically invisible and the $V_{g,n}$ remain asymptotically multiplicative.

Finally, the normalized sum of the uncorrected corner matrices is precisely the $Q_n$-block of $A_n$. Since $P_n - A_n \to 0$ and $Q_n P_n = 0$,

$$\left\| \sum_{f \in F} C_{f,n} \right\| \longrightarrow 0.$$

There are only finitely many $f \in F$, so replacing every $C_{f,n}$ by its polar correction preserves this limit and proves the lemma. □

The second ingredient is the finite-stage Kazhdan-compression calculation.

**Lemma 7.4** (compression defects collapse)**.** *Let $(V_{g,n})$ be any operator-norm asymptotic unitary representation of $H$. For a marked Kazhdan pattern $(\Lambda, \iota, t, c)$, with $d = tct^{-1}$, for every $\lambda \in \Lambda$ and $\varepsilon > 0$ there is $N$ such that, for all $n \ge N$,*

$$\left\| V_{[d,\iota(\lambda)],n} - V_{1,n} \right\|_2^2 \le \varepsilon.$$

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptCompressionDefectsCollapse`

*Proof.* Apply the adjoint microstates $\operatorname{Ad} V_{\iota(\lambda),n}$ to $L^2(M_{r_n}, \mathrm{tr}_{r_n})$. Choose a finite symmetric Kazhdan set $S \subseteq \Lambda$, with $1 \in S$, generating $\Lambda$, and a Kazhdan constant $0 < \kappa \leq 1$. Fix

$$1 - \frac{\kappa^2}{4|S|} < \theta < 1,$$

let $H_n$ be the Hermitian average of the adjoint $S$-microstates, and put $P_n = \chi_{(\theta,\infty)}(H_n)$. Spectral capture gives, for every fixed $\lambda \in \Lambda$,

$$\left\|(\operatorname{Ad} V_{\iota(\lambda),n} - 1)P_n\right\| \longrightarrow 0.$$

The constants can be fixed uniformly in the matrix size. Given a requested squared Hilbert–Schmidt error $\varepsilon > 0$, set

$$\eta = \frac{(1-\theta)\varepsilon}{192}.$$

At a sufficiently large common stage, each of the two conjugation defects of $V_{c,n}$ associated to every $s \in S$ is at most $\eta$ in operator norm. The Hermitian average therefore has residual norm at most $\eta$, and the spectral-cut inequality gives the normalized root-capture estimate

$$\frac{1}{r_n}\left\|(1-P_n)\operatorname{vec}(V_{c,n})\right\|_2^2 \leq \frac{\varepsilon}{192}. \tag{12}$$

For each generator in the Kazhdan set, (M1) supplies a $\lambda' \in \Lambda$ with $t\iota(\lambda)t^{-1} = \iota(\lambda')$. The multiplicative defects then show that the adjoint-conjugated spectral corner has asymptotically no leakage below the same cut. More precisely, if $R_n = (\operatorname{Ad} V_{t,n})P_n(\operatorname{Ad} V_{t,n})^*$, this gives

$$\|(1-R_n)P_n\| \longrightarrow 0.$$

They have equal rank at every coordinate. The finite-dimensional projection flip, applied with $\varepsilon' = \min\{1/2, \varepsilon/2\}$, bounds the reverse leakage by $\varepsilon'/\sqrt{1-(\varepsilon')^2} \leq 2\varepsilon' \leq \varepsilon$. Thus

$$\|(1-P_n)R_n\| \longrightarrow 0,$$

and hence

$$\|P_n - R_n\| \longrightarrow 0.$$

This equal-rank projection argument is the finite-dimensional replacement for the stable-finiteness step one might be tempted to perform only after taking a $C^*$-quotient; Figure 4 depicts the geometry.

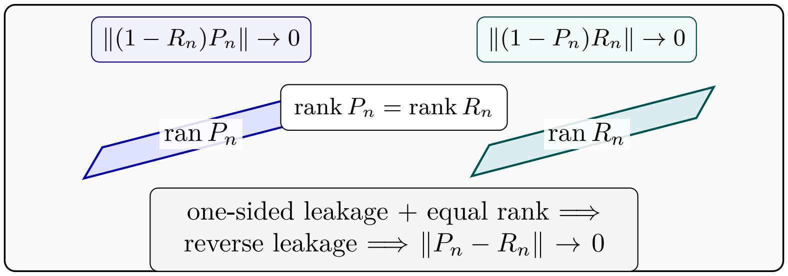


FIGURE 4. Equal-rank reversal. One-sided spectral leakage between the two schematic ranges, together with equality of their finite dimensions, forces the reverse leakage and hence convergence of the two projections.

Because $c$ centralizes $\iota(\Lambda)$, the vector represented by $V_{c,n}$ is asymptotically captured by $P_n$. The equality just obtained transports that capture to $V_{d,n}$; hence conjugation by $V_{\iota(\lambda),n}$ moves $V_{d,n}$ by a quantity tending to zero in normalized Hilbert–Schmidt norm. Unitary invariance identifies this quantity with $\left\|V_{[d,\iota(\lambda)],n} - V_{1,n}\right\|_2$, up to the vanishing multiplicative defects. More explicitly, the finite-stage vector chain gives

$$\operatorname{dist}_{2,n}([d,\iota(\lambda)],1)^2 \le 8\delta_n^2 + 64\ell_n^2 + 64\rho_n, \tag{13}$$

where $\delta_n$ is the fixed-element Kazhdan displacement, $\ell_n$ is the reverse compressor leakage, and $\rho_n$ is the normalized mass in (12). Choose their eventual bounds as

$$\delta_n \le \sqrt{\varepsilon/24}, \qquad \ell_n \le \sqrt{\varepsilon/192}, \qquad \rho_n \le \varepsilon/192.$$

Then the right side of (13) is at most

$$8\frac{\varepsilon}{24} + 64\frac{\varepsilon}{192} + 64\frac{\varepsilon}{192} = \varepsilon.$$

This proves the claim with constants independent of $r_n$. These are the same 8/64/64 finite-stage estimates used by the formal proof; no passage to an abstract limiting projection is hidden here. $\square$

## 8. PROOF OF THE CRITERION AND OF THEOREM A

*Proof of Theorem 7.2.* Suppose, for contradiction, that some corona representation $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ is nontrivial on $F$. Compose with $\kappa_{(d_n)}^{-1}$ to obtain its coordinate model. For some $f_0 \in F$ the distance from its image to 1 has positive limit superior. Pass to a subsequence on which that distance is uniformly positive and then to any free ultrafilter on the subsequence. Lemma 7.3 produces an operator-norm asymptotic unitary representation $(V_{g,n})$ such that

$$\left\| \sum_{f\in F} V_{f,n} \right\| \longrightarrow 0. \tag{14}$$

Regard the same matrices as an asymptotic representation for the normalized Hilbert–Schmidt metric, and fix a free ultrafilter $\omega$ containing the cofinite filter. They define a homomorphism $\rho_2$ into the corresponding tracial unitary ultraproduct. By Lemma 7.4,

$$\rho_2([d, \iota(\lambda)]) = 1 \qquad (\lambda \in \Lambda).$$

The kernel of a homomorphism is normal; hence it contains $N_{\mathrm{comp}}$, and therefore $F$. Consequently, for every $f \in F$,

$$\lim_{n\to\omega} \|V_{f,n} - V_{1,n}\|_2 = 0. \tag{15}$$

To return from the ultrafilter statement to an ordinary sequence, put $\eta_k = 1/(k+1)$. Since $F$ is finite, for every $k$ the set of indices $n \geq k$ for which

$$\|V_{f,n} - V_{1,n}\|_2^2 < \eta_k \qquad (f \in F)$$

belongs to $\omega$ and is therefore nonempty. Choose one such index $\phi(k) \geq k$ and reindex by $W_{g,k} = V_{g,\phi(k)}$. Cofinality of $\phi$ preserves asymptotic multiplicativity and (14), while

$$\|W_{f,k} - W_{1,k}\|_2 \longrightarrow 0 \qquad (f \in F).$$

Because $F$ is finite,

$$\left\| \sum_{f\in F} W_{f,k} - |F| W_{1,k} \right\|_2 \leq \sum_{f\in F} \|W_{f,k} - W_{1,k}\|_2 \longrightarrow 0.$$

On the other hand, $W_{1,k}$ is unitary, so $\||F|W_{1,k}\|_2 = |F|$, whereas

$$\left\| \sum_{f\in F} W_{f,k} \right\|_2 \leq \left\| \sum_{f\in F} W_{f,k} \right\| \longrightarrow 0$$

by (14) after reindexing. The triangle inequality would force $|F| = 0$, a contradiction. Thus every corona representation kills $F$. ☐

**Lemma 8.1** (the marked word is a defect square)**.** *Let $H_1$ be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$\left[\, d,\ a_1\, d\, a_1^{-1} \right] \;=\; \left[\, d, a_1 \,\right]^2.$$

**EXACT**

`GroupApproximation.commutator_conjugate_eq_commutator_sq_of_sq_eq_one`

*Proof.* Put $b = a_1 d a_1^{-1}$, so $b^2 = a_1 d^2 a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d\,(a_1 d a_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db\,db = d\,b\,d^{-1}b^{-1} = [d, b]. \qquad ☐$$

*Proof of Theorem A.* Part (1) is Proposition 4.3. For part (2), use the canonical homomorphism $\iota\colon \mathcal{B} \to E$ and put $a = v_1$. The tuple $(\mathcal{B}, \iota, t, c)$ is a marked Kazhdan pattern in $E$: Proposition 3.3 gives property (T); the six stable-letter relators give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$; and the six commutator relators give $[c, \iota(\mathcal{B})] = 1$. No injectivity of $\iota$ is required. In $E$ we have $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 8.1 with $a_1 = \iota(v_1)$ gives

$$w = [\,d,\ \iota(v_1)\,d\,\iota(v_1)^{-1}] = [\,d, \iota(v_1)\,]^2 \ \in\ N_{\mathrm{comp}},$$

since $[\,d, \iota(v_1)\,]$ is one of the defining generators of $N_{\mathrm{comp}}$. The subgroup $F = \{1, w\} = \langle w \rangle$ is finite and normal: centrality and $w^2 = 1$ are defining relations, while Proposition 4.3 gives $w \neq 1$. Moreover $F \subseteq N_{\mathrm{comp}}$. Theorem 7.2 therefore kills $w$ after composing any $\Theta\colon E \to \mathsf{U}_{\mathrm{cor}}((d_n))$ with the canonical isomorphism $\kappa_{(d_n)}$ of Lemma 6.2. Since $\kappa_{(d_n)}$ is injective, $\Theta(w) = 1$. This is part (2).

For the final assertions of Theorem A: an MF embedding of $E$ would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element $w$ to 1. If $C^*_{\mathrm{red}}(E)$ were an MF algebra, an injective, possibly nonunital $*$-homomorphism $e\colon C^*_{\mathrm{red}}(E) \to \mathcal{Q}$ would give the injective unitary homomorphism $u \mapsto e(u) + (1 - e(1))$. Composing it with the canonical map $E \to \mathcal{U}(C^*_{\mathrm{red}}(E))$, and then with $\kappa^{-1}_{(d_n)}$, would exhibit an injective corona representation of $E$: the map $g \mapsto \lambda_g$ is injective. The canonical map $E \to \mathcal{U}(C^*_{\max}(E))$ is injective as well, because coordinate evaluation at the reduced left-regular representation sends $u_g$ to $\lambda_g$. Thus an MF embedding of this concrete $C^*_{\max}(E)$ would again produce an injective corona representation of $E$ after the same $\kappa^{-1}_{(d_n)}$ conversion. $\square$

The marked radical element. The proof shows more than non-injectivity of each individual $\Theta$: the fixed element $w$ lies in the kernel of *every* corona representation of $E$, so every corona representation factors through $E/\langle w \rangle$ (note $\langle w \rangle = \{1, w\}$ is central, hence normal). In the language of approximation radicals, $w$ is a nontrivial element of the "MF radical" of $E$. This terminology is meant in analogy with Slofstra's earlier quotients $G^{\mathrm{fin}}$ and $G^{\mathrm{fa}}$, obtained by killing elements invisible to exact and approximate finite-dimensional representations, respectively [Slo17, Definitions 2.5–2.7]. By Theorem 7.2 the same holds in every countable group with a marked Kazhdan pattern: every finite normal subgroup of the compression-defect subgroup lies in the MF radical. Section 9 develops the radical systematically; we do not know whether $E/\langle w \rangle$ is MF, i.e. whether the MF radical of $E$ is exactly $\{1, w\}$.
No density hypothesis. No lower bound on the corner ranks $r_n$, or on their density in $d_n$, is used anywhere: Lemma 7.3 equips each nonzero complementary block with its own normalized trace. Thus all subsequent Hilbert–Schmidt estimates are relative to that block. This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 11).

## 9. Consequences

Throughout this section, "corona representation" and "MF" are as in Section 2.

We record the universal MF radical, the reduced-group-$C^*$ consequence, and the failure of quotient permanence. These are direct consequences of the same literal marked element; no additional envelope or recursion-theoretic construction is used.

### 9.1. **The MF radical.**

**Definition 9.1.** For a countable group $G_1$, the *MF radical* is

$$\mathrm{Rad}_{\mathrm{MF}}(G_1) \;=\; \bigcap\{\ker\Theta : \;\Theta \text{ a corona representation of } G_1\},$$

a normal subgroup of $G_1$. This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the approximation kernels studied in [NST].

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptMFRadical`

**Lemma 9.2** (portability)**.** *For every homomorphism $\varphi\colon G_1 \to G_2$ of countable groups,*

$$\varphi(\mathrm{Rad}_{\mathrm{MF}}(G_1)) \;\subseteq\; \mathrm{Rad}_{\mathrm{MF}}(G_2) :$$

*the MF radical is functorial. In particular, if $\varphi\colon E \to G_2$ satisfies $\varphi(w) \neq 1$, then $G_2$ is not MF.*

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptRadicalPortability`

*Proof.* If $x \in \mathrm{Rad}_{\mathrm{MF}}(G_1)$ and $\Theta$ is a corona representation of $G_2$, then $\Theta \circ \varphi$ is a corona representation of $G_1$, so $\Theta(\varphi(x)) = 1$. For the second statement: $w \in \mathrm{Rad}_{\mathrm{MF}}(E)$ by Theorem A, so $1 \neq \varphi(w) \in \mathrm{Rad}_{\mathrm{MF}}(G_2)$, and an injective corona representation of $G_2$ would kill $\varphi(w)$, a contradiction. ☐

**Proposition 9.3** (the largest MF quotient)**.** *Let $G_1$ be a countable group and $R = \mathrm{Rad}_{\mathrm{MF}}(G_1)$.*

*(1) $R$ is the kernel of a single corona representation of $G_1$.*
*(2) $G_1/R$ is MF.*
*(3) Every homomorphism from $G_1$ to an MF group factors through $G_1/R$.*

*In particular $G_1$ is MF if and only if $R = \{1\}$. The statement is presumably folklore once the radical is defined; we include a proof for completeness and claim no novelty for it.*

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptLargestMFQuotient`

*Proof.* (3) If $\varphi\colon G_1 \to M$ with $M$ MF and $j\colon M \to \mathsf{U}_{\mathrm{cor}}((d_n))$ is an injective corona representation, then $j \circ \varphi$ is a corona representation of $G_1$, so $j(\varphi(R)) = 1$ and, by injectivity of $j$, $\varphi(R) = 1$.

(1) and (2). If $R = G_1$, the trivial representation has kernel $R$ and $G_1/R$ is trivial, hence MF; so assume $R \neq G_1$ and let $(g_i)_{i\geq 1}$ list, possibly with repetitions, elements of $G_1 \setminus R$ meeting every nontrivial coset of $R$. For each

$i$ choose a corona representation and write it, through $\kappa_{(d_n^{(i)})}$, as $\Theta_i\colon G_1 \to \mathcal{U}(\mathcal{Q}_i)$, $\mathcal{Q}_i = \mathcal{Q}((d_n^{(i)})_{n\in\mathbb{N}})$, with $\Theta_i(g_i) \neq 1$; put $\varepsilon_i = \frac{1}{2}\,\|\Theta_i(g_i) - 1\| > 0$, and choose unitary lifts $(U_{g,n}^{(i)})_{n\in\mathbb{N}}$ of $\Theta_i(g)$ for every $g \in G_1$ (Lemma 6.1), with $U_{1,n}^{(i)} = 1$. Recall that the quotient norm in a corona is the limit superior of the coordinate norms: subtracting a null sequence does not change the limit superior, and truncating finitely many coordinates realizes it.

Enumerate $G_1 = \{h_1, h_2, \dots\}$ and fix a stage $k \geq 1$. For each $i \leq k$: the finitely many products $U_{g,n}^{(i)} U_{h,n}^{(i)}$ and $U_{gh,n}^{(i)}$ with $g, h \in \{h_1, \dots, h_k\}$ lift the same corona elements, so there is $n_0$ with all the corresponding defects at most $1/k$ for $n \geq n_0$; and since $\limsup_{n\to\infty} \left\| U_{g_i,n}^{(i)} - 1 \right\| = 2\varepsilon_i$, there are infinitely many $n \in \mathbb{N}$ with $\left\| U_{g_i,n}^{(i)} - 1 \right\| \geq \varepsilon_i$. Choose $n_i(k) \in \mathbb{N}$ satisfying both conditions and set

$$\Phi_k(g) \;=\; \bigoplus_{i\leq k} U_{g,\,n_i(k)}^{(i)} \;\in\; \mathcal{U}(M_{D_k}(\mathbb{C})), \qquad D_k = \sum_{i\leq k} d_{n_i(k)}^{(i)}.$$

Define $\Theta_\infty(g)$ as the class of $(\Phi_k(g))_{k\in\mathbb{N}}$ in $\mathsf{U}_{\mathrm{cor}}((D_k)_{k\in\mathbb{N}})$. For fixed $g, h \in G_1$, both lie in $\{h_1, \dots, h_k\}$ for all large $k$, and the operator norm of a block sum is the maximum of the block norms, so $\|\Phi_k(g)\Phi_k(h) - \Phi_k(gh)\| \leq 1/k \to 0$; together with $\Phi_k(1) = 1$ this makes $\Theta_\infty$ a corona representation of $G_1$. Consequently $\Theta_\infty(R) = 1$. Conversely, if $g \notin R$, pick $i$ with $g \in g_i R$; then $\Theta_\infty(g) = \Theta_\infty(g_i)$, and for every $k \geq i$, $\|\Phi_k(g_i) - 1\| \geq \left\| U_{g_i,n_i(k)}^{(i)} - 1 \right\| \geq \varepsilon_i$, so the representative of $\Theta_\infty(g_i)$ stays at least $\varepsilon_i$ from the identity and $\Theta_\infty(g) \neq 1$. Hence $\ker \Theta_\infty = R$, proving (1), and $\Theta_\infty$ descends to an injective corona representation of $G_1/R$, proving (2). For the final equivalence: if $G_1$ is MF, an injective corona representation forces $R = \{1\}$; if $R = \{1\}$, then (2) says $G_1 = G_1/R$ is MF. $\square$

**Corollary 9.4** (exact radical from a candidate quotient)**.** *Let $N \trianglelefteq G_1$ satisfy $N \leq \mathrm{Rad}_{\mathrm{MF}}(G_1)$. Every corona representation of $G_1$ factors uniquely through $G_1/N$. If in addition $G_1/N$ is MF, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G_1) = N.$$

*Thus proving that a concrete defect subgroup is invisible and that its quotient is MF computes both the entire MF radical and the largest MF quotient.*

**EXACT**

`GroupApproximation.ManuscriptExactWrappers.manuscriptExactRadicalFromCandidateQuotient`

*Proof.* The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every corona representation $\Theta$. For the reverse containment in the displayed equality, compose the quotient map $G_1 \to G_1/N$ with an injective corona representation of $G_1/N$: its kernel is exactly $N$, while the MF radical lies in the kernel of every corona representation. $\square$

**Corollary 9.5.** *Let $A$ be a unital $C^*$-algebra admitting an injective, possibly nonunital, $*$-homomorphism into a norm matrix corona. Then the group $E$ admits no injective homomorphism into $\mathcal{U}(A)$.*

EXACT

`GroupApproximation.LiteralNonMFConsequences.literal_no_faithful_corona_subalgebra_target`

*Proof.* Let $j\colon A \to Q$ be the possibly nonunital inclusion into the ambient corona and let $\phi\colon E \to \mathcal{U}(A)$. Complement correction gives

$$\widehat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(Q).$$

This is an injective homomorphism on $\mathcal{U}(A)$, so $\widehat{j} \circ \phi$ is a corona representation of $E$. The exact literal kernel theorem sends $w$ to the identity there; injectivity of $\widehat{j}$ gives $\phi(w) = 1$. Proposition 4.3 has $w \neq 1$, so $\phi$ cannot be injective. □

9.2. **Stable finiteness does not imply MF.**

**Lemma 9.6.** *Let $G_1$ be a countable discrete group.*

*(1) The canonical tracial state $\tau$ on $C^*_{\mathrm{red}}(G_1)$ is faithful.*
*(2) Every unital $C^*$-algebra $A$ with a faithful tracial state is stably finite: for every nonempty finite index set $I$, every isometry in $M_I(A)$ is a unitary. In particular this holds in $M_k(A)$ for every $k \geq 1$.*

EXACT

`GroupApproximation.ManuscriptExactWrappers.manuscriptFaithfulTraceAndStableFiniteness`

*Proof.* (1) Realize $C^*_{\mathrm{red}}(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries $\lambda_g$, and let $\rho_g$ be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$; each $\rho_g$ commutes with each $\lambda_{g'}$, hence with every element of $C^*_{\mathrm{red}}(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e \rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}} x\delta_e = 0$ for every $g \in G_1$. The vectors $\delta_g$ are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k}\sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k}\sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the $i$-th diagonal entry of $x^*x$ is $\sum_j x^*_{ji}x_{ji}$, so $\tau_k(x^*x) = 0$ forces $\tau(x^*_{ji}x_{ji}) = 0$, hence $x_{ji} = 0$, for all $i, j$. If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. □

*Proof of Theorem D.* $C^*_{\mathrm{red}}(E)$ is unital and separable since $E$ is countable; by Lemma 9.6 it carries a faithful tracial state and is stably finite. It is not an MF algebra by Theorem A. □

### 9.3. **Failure of quotient permanence.**

**Corollary 9.7** (quotients)**.** *MF groups are not closed under quotients: the free group $F_8$ is MF, and the literal group $E$ is its quotient but is not MF.*

EXACT

`GroupApproximation.LiteralMFQuotientControls.manuscriptQuotientNonclosure`

*Proof.* Map the eight free generators onto the eight displayed generators of $E$. The resulting homomorphism is surjective because those generators define the literal presentation. The group $F_8$ is residually finite and hence MF, while Theorem A shows that $E$ is not. $\square$

## 10. The cyclic-base comparison

We prove Theorem C for the literal seven-relator presentation. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and $c$ centralizes this cyclic base. The same commutant argument as in Theorem B therefore maps $w_{\mathrm{BS}}$ to the identity in every finite-dimensional representation over every field.

For nontriviality, use the affine realization of $\mathrm{BS}(1,2)$ and its coset action. The root site is fixed by the cyclic base, while the two sites represented by $t$ and $\gamma_0 t$ are distinct. Attach Clifford generators to these sites and map $c$ to the root lamp. All seven relators hold, and

$$w_{\mathrm{BS}} \longmapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\mathrm{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjective realized quotient. This comparison isolates the precise limitation of the finite-dimensional argument: exact finite-dimensional invisibility does not itself imply that the marked word is trivial.

## 11. Limitations: Hilbert–Schmidt metrics and hyperlinearity

The proof of Theorem A does not begin from a merely tracial approximation, for a structural reason worth recording.

Suppose one replaces the norm corona $\mathcal{Q}$ by a tracial ultraproduct $\mathcal{M} = \prod^{\omega}(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$, with defects measured in $\|\cdot\|_2$ rather than operator norm, and asks whether a hyperlinear-style embedding of $E$ must kill $w$. The first coordinate step already breaks: Hilbert–Schmidt multiplicative defects do not control the operator norms of the off-diagonal blocks cut out by the Reynolds spectral projections. Consequently principal compression need not remain multiplicative in operator norm, its Gram defects need not vanish in operator norm, and the dimension-free polar-repair estimate used in Lemma 7.3 is unavailable. Likewise, the finite-stage Kazhdan argument needs operator-norm spectral leakage estimates before it passes to normalized Hilbert–Schmidt norm. A tracial ultraproduct retains only the latter information, so the equal-rank coordinate flip cannot be invoked. The present argument therefore gives no universal tracial analogue of the MF theorem and decides neither hyperlinearity of $E$ nor the sofic-versus-hyperlinear problem.

The dividing line in the present argument is thus: *operator-norm matrix models form a stably finite world, in which property (T) compression is rigid; for tracial matrix models the available ambient algebra of the conjugation action can be properly infinite, so the same projection comparison need not collapse to equality.*

This failure is closely related to a current structural question rather than to a known impossibility theorem. Alekseev–Thom ask whether, for a Kazhdan representation into a tracial ultraproduct of matrices, its commutant can be recovered as an ultraproduct of finite-dimensional subalgebras, possibly as coordinate centralizers of suitable lifts [AT, Open Problem 6.2]. Such a coordinate model would provide finite-level structure absent from the argument above, but their question is open. Some additional hypothesis is therefore essential before the operator-norm proof can be transferred to a tracial setting.

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